A. The function has a point where behavior might change

Examples:

- Piecewise functions
- Absolute value function
- Rational functions with denominator zero
- Functions with holes/jumps/corners

If the question specifically asks: **"Does lim(x → a) f(x) exist?"** you must check both sides.

B. When you suspect a one-sided mismatch

Examples:

- Graph changes suddenly at a point
- The limit involves |x|, sign changes, etc.

When you do NOT need both sides

When the function is known to be **smooth and well-behaved** (polynomial, exponential, trigonometric, or rational where denominator ≠ 0). Substitution works unless it creates an indeterminate form.

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3. When do we take approximate numbers like 0.999 or 1.001?

You use values like **0.999** or **1.001** when:

A. Understanding behavior numerically

Useful when:

- Graphing
- Making a table of values
- Handling limits that give tricky forms like 0/0

For example, checking the limit as x → 1:

- 0.999 → left side
- 1.001 → right side

B. When checking one-sided limits numerically

- Left-hand limit → **a − ε** (e.g., 0.999)
- Right-hand limit → **a + ε** (e.g., 1.001)

C. When the limit is difficult to visualize

Using approximate numbers helps build intuition.

Category: This is part of the **numerical/table-based limit method**, not a separate type of limit.

Summary

- A limit is continuous when both sides exist, are equal, and equal to the function value.
- Check both sides when left and right behavior might differ.
- Use 0.999 or 1.001 when exploring numerical/graphical or one-sided limit behavior.